

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

QIAN QI

ABSTRACT. We formulate the induced empirical process at deterministic inducing-block speed. The state retains vertices, admissible edge flow, return and roof marks, and actual excursion path profiles. Finite-core Markov large deviations pass through a projective exponential-tightness theorem, while an actual-excursion recession functional and terminal-profile clock surgery give full collision- and physical-time empirical-path large deviations.

Platform identifier: TL3-LPATH-v3. **Controlling revision:** ROUND10-REFEREE-POSITIVE-CLOSURE.
Registered proof source: A3_DETERMINISTIC_SPEED_PROJECTIVE_FLOW_LDP.tex.

CONTENTS

1. Round-ten reconstruction: deterministic-speed marked flows and a projective renewal LDP	1
1.1. The deterministic block-flow state	2
1.2. Finite cores and exact contractions	2
1.3. Recession as many actual excursions	3
1.4. Collision and physical deterministic horizons	3
Scope and verification status	4

1. ROUND-TEN RECONSTRUCTION: DETERMINISTIC-SPEED MARKED FLOWS AND A PROJECTIVE RENEWAL LDP

Let $\mathcal{A}$ be the countable first-return alphabet of a fixed Young base, with admissible edge set $\mathcal{E} \subset \mathcal{A}^2$, return length $r(a)$, roof $\tau(a)$, Gibbs potential $\phi(a, b)$, and actual excursion path profile $K(y, a)$. Branch labels and transition edges are retained in the state; a branch average is never substituted for the realized excursion.

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, Young towers, level-3 large deviations, empirical path measures, information projection.

1.1. The deterministic block-flow state. For the first N complete excursions define

$$L_N^e = \frac{1}{N} \sum_{j=0}^{N-1} \delta_{(a_j, a_{j+1}, K(y_j, a_j), r(a_j), \tau(a_j))}.$$

The speed is the deterministic integer N . Let $\mathfrak{E}$ be the Polish space of probability measures on the marked edge space with the weak topology plus convergence of the weight $1 + r + \tau + |\phi|$. Its stationary subset has equal source and target alphabet marginals.

Lemma 1.1 (State sufficiency and exponential tightness). *The entropy rate, Gibbs-potential average, collision clock, physical clock, transition admissibility and realized excursion-path barycenter are Borel functions of L_N^e . Moreover (L_N^e) is exponentially tight at speed N in $\mathfrak{E}$.*

Proof. The source and target branch labels are explicit coordinates, so the conditional transition kernel and its entropy are recovered by disintegration. The remaining quantities are coordinate integrals. The Young return tail and finite-horizon bound give an exponential moment of $r + \tau$; summable variations control the negative part of ϕ . Chernoff bounds give uniform integrability of the weight. Tightness of finite-window path profiles follows from the singularity shield. Weighted Prokhorov compactness gives exponential tightness. $\square$

1.2. Finite cores and exact contractions. Let $\mathcal{A}_L$ contain the first L branches together with a fixed finite connector set. Write π_L for the projection which keeps all edges inside $\mathcal{A}_L$ and sends the complement to a marked cemetery state without forgetting its total weight.

Theorem 1.2 (Finite-core marked-flow LDP). *For every L , $\pi_L L_N^e$ satisfies a full LDP at speed N with good rate*

$$I_L(\lambda) = \begin{cases} D(q_\lambda \| q_\phi \mid \lambda_1), & \lambda \text{ stationary and supported on admissible edges,} \\ +\infty, & \text{otherwise,} \end{cases}$$

including the potential normalization and retained cemetery weight. If $L < L'$, then

$$I_L(\lambda) = \inf_{\pi_{L,L'} \lambda' = \lambda} I_{L'}(\lambda').$$

Proof. On the finite core, the transfer matrix with cylinder sources is a positive finite matrix. Perron–Frobenius perturbation gives its log moment generating function on all finite sources. Exponential tilting gives the lower bound and finite-state entropy duality gives the rate. The cemetery mark records every discarded edge and its weight, so projection from L' to L is an actual continuous contraction of the same empirical flow. The contraction principle proves exact consistency of the finite rates. $\square$

Theorem 1.3 (Projective marked-flow LDP). *The deterministic-speed flows L_N^e satisfy a full good LDP on $\mathfrak{E}$ with*

$$I_e(\lambda) = \sup_L I_L(\pi_L \lambda).$$

Every finite-rate λ admits finite-core stationary recovery flows using only edges in $\mathcal{E}$, with convergence of entropy, potential, clock moments and realized path profiles.

Proof. Dawson–Gärtner gives the projective weak LDP. Exponential tightness from Lemma 1.1 upgrades it to the weighted topology and makes the rate good. For recovery, truncate a finite-rate stationary edge flow where its weighted tail is small. Balance lost in/out mass along the fixed legal connector network by a finite circulation. Mixing with a small positive circulation puts the retained kernel in the relative interior without introducing forbidden edges. Conditional entropy converges by uniform integrability of its information density; profiles and clocks converge by weighted truncation. $\square$

1.3. Recession as many actual excursions. For $M \geq 1$, let $\mathcal{R}_M$ be the empirical law of M consecutive actual excursions conditioned to have total collision clock at least L . Define the finite-window logarithmic costs and take their lower-semicontinuous projective epigraph limit, first as $L \rightarrow \infty$, then as $M \rightarrow \infty$. Call the result J_∞ .

Theorem 1.4 (Actual recession principle). *The rate J_∞ exists and is convex. Every sequence of actual long-excursion blocks obeys its Gamma-liminf bound, and every finite-cost recession law has a recovery sequence made of legal consecutive excursions and the fixed connector network. No convex mixture is asserted to be one first-return branch.*

Proof. At a fixed finite path window, log probabilities are subadditive under concatenation of blocks of complete excursions; visits to the Young base are allowed between excursions and are precisely the renewal delimiters. The connector network has bounded clock and bounded logarithmic Gibbs cost, so it contributes $o(L)$. Fekete’s lemma gives the finite-window limits. Projective consistency and exponential clock tightness give a unique lower-semicontinuous epigraph limit. Concatenating proportions of two long blocks proves convexity. A diagonal choice over path windows yields actual recovery blocks and both Gamma bounds. $\square$

1.4. Collision and physical deterministic horizons. For a deterministic collision horizon n , stop at the first excursion crossing n and retain the cut terminal excursion. Define the analogous object at deterministic physical time T , cutting the last suspension segment at exactly T .

Lemma 1.5 (Clock inversion with a retained terminal profile). *On every rate sublevel, the complete-excursion count divided by n , respectively by T , is exponentially equivalent to the reciprocal mean collision, respectively physical, clock. The terminal cut contributes at most one excursion and*

its normalized path mass is exponentially negligible; it nevertheless hits the prescribed physical time exactly.

Proof. Exponential moments of r and τ control the maximum excursion among linearly many renewals. A terminal excursion of macroscopic size carries the recession cost already represented by Theorem 1.4; otherwise its normalized mass vanishes. Renewal inversion gives the deterministic clock ratios. Cutting a suspension leg changes neither its preceding collision profile nor more than $o(T)$ path time. $\square$

Theorem 1.6 (Full collision- and physical-time empirical-path LDP). *The collision-time empirical path law at deterministic horizon n satisfies a good LDP at speed n , and the physical-time law at deterministic horizon T satisfies a good LDP at speed T . Their rates are the contractions of*

$$I_e(\lambda) + J_\infty(\zeta)$$

under exact recurrent/recession clock normalization and the realized-path barycenter. The upper and lower bounds include all finite-rate recurrent and recession phases.

Proof. Combine the deterministic block-flow LDP with Theorem 1.4. The joint state is exponentially tight because escaped clock mass is recorded explicitly. Lemma 1.5 makes collision and physical inversion an exponentially good map. The extended contraction principle gives the upper bound. For the lower bound, concatenate a finite-core recurrent recovery flow with an actual recession recovery block in prescribed proportions, use legal connectors, and cut only the terminal excursion. Connector and cut costs are sublinear. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn